

ocaml checklist

OCaml Exercises Checklist

Lists

- ☐ Tail of a List
- ☐ Last Two Elements of a List
- ☐ N'th Element of a List
- ☐ Length of a List
- ☐ Reverse a List
- ☐ Palindrome
- ☐ Flatten a List
- ☐ Eliminate Duplicates
- ☐ Pack Consecutive Duplicates
- ☐ Run-Length Encoding
- ☐ Modified Run-Length Encoding
- ☐ Decode a Run-Length Encoded List
- ☐ Run-Length Encoding of a List (Direct Solution)
- ☐ Duplicate the Elements of a List
- ☐ Replicate the Elements of a List a Given Number of Times
- ☐ Drop Every N'th Element From a List
- ☐ Split a List Into Two Parts
- ☐ Extract a Slice From a List
- ☐ Rotate a List N Places to the Left
- ☐ Remove the K'th Element From a List
- ☐ Insert an Element at a Given Position Into a List
- ☐ Create a List Containing All Integers Within a Given Range
- ☐ Extract Randomly Selected Elements From a List
- ☐ Lotto: Draw N Different Random Numbers From 1..M
- ☐ Generate a Random Permutation of a List
- ☐ Generate Combinations of K Distinct Objects
- ☐ Group the Elements of a Set Into Disjoint Subsets
- ☐ Sort a List of Lists According to Length



Arithmetic

- ☐ Determine Whether a Number Is Prime
 - ☐ Greatest Common Divisor (GCD)
 - ☐ Determine Whether Two Numbers Are Coprime
 - ☐ Euler's Totient Function $\phi(m)$
 - ☐ Determine the Prime Factors of a Number
 - ☐ Determine the Prime Factors (Improved Version)
 - ☐ Euler's Totient Function (Improved)
 - ☐ Compare Totient Implementations
 - ☐ Generate a List of Prime Numbers
 - ☐ Goldbach's Conjecture
 - ☐ List of Goldbach Compositions
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Logic and Codes

- ☐ Truth Tables (2 Variables)
 - ☐ Truth Tables (General Case)
 - ☐ Gray Code
 - ☐ Huffman Code
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Binary Trees

- ☐ Construct Completely Balanced Binary Trees
- ☐ Symmetric Binary Trees
- ☐ Binary Search Trees

- ☐ Generate-and-Test Paradigm
- ☐ Construct Height-Balanced Binary Trees
- ☐ Height-Balanced Trees With Given Nodes
- ☐ Count the Leaves of a Binary Tree
- ☐ Collect Leaves in a List
- ☐ Collect Internal Nodes in a List
- ☐ Collect Nodes at a Given Level
- ☐ Construct a Complete Binary Tree
- ☐ Layout a Binary Tree (1)
- ☐ Layout a Binary Tree (2)
- ☐ Layout a Binary Tree (3)
- ☐ String Representation of Binary Trees
- ☐ Preorder and Inorder Sequences
- ☐ Dotstring Representation



Multiway Trees

- ☐ Tree Construction From a Node String
- ☐ Count the Nodes of a Multiway Tree
- ☐ Internal Path Length
- ☐ Bottom-Up Order Sequence
- ☐ Lisp-Like Tree Representation



Graphs

- ☐ Graph Conversions
- ☐ Path From One Node to Another
- ☐ Cycle From a Given Node

- ☐ Construct All Spanning Trees
- ☐ Minimal Spanning Tree
- ☐ Graph Isomorphism
- ☐ Node Degree and Graph Coloring
- ☐ Depth-First Traversal
- ☐ Connected Components
- ☐ Bipartite Graphs
- ☐ Generate K-Regular Simple Graphs



Miscellaneous

- ☐ Eight Queens Problem
- ☐ Knight's Tour
- ☐ Von Koch's Conjecture
- ☐ Arithmetic Puzzle
- ☐ English Number Words
- ☐ Syntax Checker
- ☐ Sudoku
- ☐ Nonograms
- ☐ Crossword Puzzle

